

The Effect of NHL Expansion on Player Salaries

Jude Yurcich

Introduction

- NHL labor structure is very unique
 - Entirely unionized with long contracts and guaranteed money (Jones 1969)
- Today, there are 32 teams with 23 NHL players each
- However, expansion could change the dynamic of NHL labor entirely
- What is the impact of NHL expansion on NHL players' salaries?

Literature Review

- Economics of NHL are very unique
 - “Work performance” can be entirely quantified with sports stats
 - NHL teams have strong financial wellbeing in Canada and less so in the US (Jones and Ferguson 1988)
- Financial performance of the league is driven by hedonic principles
 - Hedonic motivations bring fans to spend money on teams by way of games and other means of spending on sports (Davis and Miller 2023)
 - Humphreys and Mondello use hedonic motivation to find that team valuations increased by 15% from 1969 (2008)
- Determination of player salaries comes from a multitude of factors
 - No one agreed upon model exists to determine player salaries or performance (Dixon 2003)
 - A significant causal relationship exists between marginal tax rate and gross salary (Beirnes)
- Hypothesis: NHL expansion will increase players’ salaries due to a general trajectory of financial strength and increased competition for players’ services

Data

- Data comes from multiple sources
 - Hockey Reference contains all on ice statistics, Spotrac contains all cap statistics, and inflation statistics come from the Minneapolis Federal Reserve
- Combination of these statistics creates a picture of the financial health of the league, the performance of the players, and the general picture of the economy
- Panel data with over 300 data points for each team for each year they were in the league

Data

	(1)			
	Summary Statistics			
	mean	sd	min	max
AdjustedSalaries	1.07e+08	2.14e+07	6.92e+07	1.69e+08
Attendance	629705.6	221612.8	0	932098
HomeGoals	122.2212	21.0045	53	192
CapSpace	2847536	5249782	-8475808	2.25e+07
LTIR	7336969	4838056	465402	2.55e+07
LogSalaries	18.46484	.1953496	18.05323	18.94375
N	312			

Sources: Hockey
Reference, Minneapolis
Federal Reserve, Spotrac

Model

OLS: $\log\text{ofadjustedsalaries}_{it} = \beta_0 + \beta_1 \text{sumatt} + \beta_2 \text{sumhomegoals} + \beta_3 \text{capspaceall} + \beta_4 \text{injuredlongterm} + \beta_5 \text{distance} + \beta_6 \text{canada} + \beta_7 \text{playoff} + \beta_8 \text{stanleycup} + \beta_9 \text{post2018} + \beta_{10} \text{post2021} + \epsilon$

FE: $\log\text{ofadjustedsalaries}_{it} = \beta_0 + \beta_1 \text{sumatt} + \beta_2 \text{sumhomegoals} + \beta_3 \text{capspaceall} + \beta_4 \text{injuredlongterm} + \beta_5 \text{distance} + \beta_6 \text{canada} + \beta_7 \text{playoff} + \beta_8 \text{stanleycup} + \beta_9 \text{post2018} + \beta_{10} \text{post2021} + \text{teamfixedeffects} + \epsilon$

Results

OLS Regression Results

	(1)	(2)	(3)	(4)	(5)
	LogSalaries	LogSalaries	LogSalaries	LogSalaries	LogSalaries
post2018	0.043** (0.018)	0.055*** (0.019)	0.056*** (0.019)	0.110** (0.054)	0.124** (0.056)
post2022	0.344*** (0.020)	0.342*** (0.020)	0.329*** (0.020)	0.339*** (0.079)	0.322*** (0.088)
Attendance		0.000* (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
HomeGoals			0.002*** (0.000)	0.001 (0.002)	0.002 (0.002)
LogCapSpace				-0.026* (0.015)	-0.028* (0.015)
LogLTIR				0.007 (0.030)	0.016 (0.033)
Distance					-0.091 (0.058)
CanadaTeam					0.004 (0.058)
PlayoffTeam					0.032 (0.053)
StanleyCup					0.000 (.)
_cons	18.342*** (0.016)	18.296*** (0.029)	18.158*** (0.042)	18.409*** (0.530)	18.275*** (0.590)
r2_a	0.575	0.578	0.594	0.580	0.572

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Results

FE Regression Results

	(1)	(2)	(3)	(4)	(5)
	LogSalaries	LogSalaries	LogSalaries	LogSalaries	LogSalaries
post2018	0.041** (0.016)	0.042** (0.018)	0.043** (0.017)	0.015 (0.050)	0.012 (0.054)
post2022	0.341*** (0.017)	0.340*** (0.017)	0.330*** (0.018)	0.577*** (0.074)	0.574*** (0.080)
Attendance		0.000 (0.000)	-0.000 (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
HomeGoals			0.001** (0.000)	0.005** (0.002)	0.005** (0.002)
LogCapSpace				-0.006 (0.016)	-0.007 (0.018)
LogLTIR				-0.067 (0.039)	-0.066 (0.042)
Distance					0.000 (.)
CanadaTeam					0.000 (.)
PlayoffTeam					-0.018 (0.063)
StanleyCup					0.000 (.)
_cons	18.344*** (0.012)	18.339*** (0.028)	18.235*** (0.049)	19.361*** (0.597)	19.372*** (0.641)
r2_a	0.586	0.584	0.593	0.782	0.749

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Results

- Similar effects for OLS and FE regressions
- Expansion tends to increase player salaries overall by around 18.3%
- Significantly higher coefficient for post-2022 than post-2018 variable
- Overall regression results suggest slight causal relationship between expansion and increased player salaries
- Limitations: Multicollinearity, OVB, small data set

Conclusion

- “Cracking the code” of how players’ salaries are determined in the NHL could be huge for both the owners and players’ union
- This study is a good insight into how labor economics works at an angle more palatable than other studies
- Would more data or a different model show a stronger or weaker relationship?
- Future avenues of research include expansion’s other financial impacts and a labor economics-based study of the modern NHL

References

Beirnes, S. The Impact of Income Tax Rates on Gross Salaries for Highly Skilled and Mobile Agents.

Dixon, Jess Charles. “Salary Determination in the National Hockey League: The Development of an Equitable Player Compensation Model.”

Handle.net, 2003, hdl.handle.net/20.500.14776/3059. Accessed 24 Oct. 2025.

Federal Reserve Bank of Minneapolis. “Consumer Price Index, 1913- | Federal Reserve Bank of Minneapolis.” *Federal Reserve Bank of*

Minneapolis, 2023, www.minneapolisfed.org/about-us/monetary-policy/inflation-calculator/consumer-price-index-1913-.

Humphreys, B. R., & Mondello, M. (2008). Determinants of Franchise Values in North American Professional Sports Leagues: Evidence from a Hedonic Price Model. *International Journal of Sport Finance*, 3(2), 98-105. <https://doi.org/10.1177/155862350800300202> (Original work published 2008)

Jones, J. C. H. (1969). The Economics of the National Hockey League. *The Canadian Journal of Economics / Revue Canadienne d'Economie*, 2(1), 1–20.

Jones, J. C. H., & Ferguson, D. G. (1988). Location and Survival in the National Hockey League. *The Journal of Industrial Economics*, 36(4), 443–457. <https://doi.org/10.2307/2098449>

References

- Miller, J. J., & Davis, M. A. (2023). An Analysis of Violence and Performance Factors' Influence on National Hockey League Attendance: A Conceptual Analysis of Hedonic Consumption. *Journal of Sport Behavior*, 46(3), 39+.
- https://link-gale-com.rlib.pace.edu/apps/doc/A791102879/AONE?u=nysl_me_pace&sid=ebsco&xid=b92799ac
- “NHL Stats, History, Scores, & Records | Hockey-Reference.com.” *Hockey-Reference.com*, Sports Reference, 2000, www.hockey-reference.com/. Accessed 1 Dec. 2025.
- spotrac. “Spotrac.com.” *Sports Contracts, Salaries, Caps, Bonuses, & Transactions*, 2019, www.spotrac.com/. Accessed 1 Dec. 2025.

Thank you for listening!